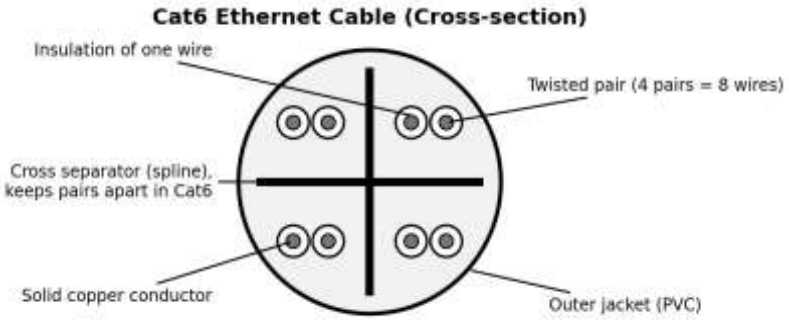
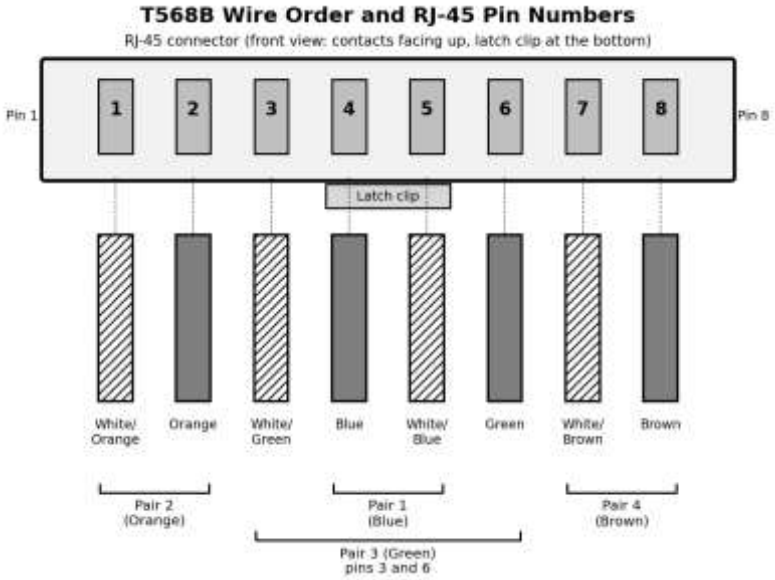
Network Cables and Connectors

Q1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



A Cat6 cable has 4 twisted pairs (8 copper wires) inside one outer jacket. Twisting each pair reduces crosstalk (interference between wires), and the cross-shaped separator (spline) in Cat6 keeps the pairs apart, which gives better speed and less noise than Cat5e.



Pin	Wire Colour (T568B)	Pair	Use (10/100 Mbps Ethernet)
1	White/Orange	Pair 2	Transmit data (TX+)
2	Orange	Pair 2	Transmit data (TX-)
3	White/Green	Pair 3	Receive data (RX+)
4	Blue	Pair 1	Not used in 10/100 Mbps (used in Gigabit and PoE)
5	White/Blue	Pair 1	Not used in 10/100 Mbps (used in Gigabit and PoE)
6	Green	Pair 3	Receive data (RX-)
7	White/Brown	Pair 4	Not used in 10/100 Mbps (used in Gigabit and PoE)
8	Brown	Pair 4	Not used in 10/100 Mbps (used in Gigabit and PoE)

Q2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Cable	Real-world use	Why it is used there
Cat5e (Twisted pair)	Zomato POS terminal / billing counter at a restaurant, and a home WiFi router to PC link	It supports speeds up to 1 Gbps for up to 100 m, which is enough for billing and order data. It is cheap and easy to install, so it suits small, low-cost setups.
Cat6 (Twisted pair)	Gaming consoles and PCs in a gaming zone, and office switch-to-computer links	It has less crosstalk because of its spline and thicker wires. It gives stable 1 Gbps (and 10 Gbps for up to 55 m), which is good for online gaming with low delay.
Fiber optic	IPL streaming server / data centre links, and ISP connections like Jio Fiber to the home	It sends data as light, so it gives very high speed over very long distances (many km) and is not affected by electrical interference. This is needed for streaming live video to lakhs of viewers.

Q3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Video referred to: (https://www.youtube.com/watch?v=N1dcCzfgdhs)

Things needed: Cat6 cable, RJ-45 connectors, cable stripper, crimping tool, wire cutter and a cable tester.

1. Cut the end of the cable straight so that the end is clean and even.
2. Remove about 2.5 to 3 cm of the outer jacket using the stripper. Be careful not to cut the thin wire insulation inside.
3. Separate the 4 twisted pairs and untwist them. Cut off the plastic cross separator (spline) if it is present.
4. Straighten the 8 wires and arrange them side by side in the T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5. Hold the wires flat and tight, and cut the tips with the cutter so that all wires are of the same length, about 1.2 cm outside the jacket.
6. Hold the RJ-45 connector with the clip facing down. Slide the wires in gently until each wire touches the front end of the connector. The outer jacket should go a little inside the connector so the plug can hold it.
7. Look at the connector from the top to check that the colour order is correct and the copper tips reach the end.
8. Put the connector in the crimping tool and squeeze firmly until it is fully pressed. The metal pins cut through the wire insulation and the connector grips the cable.
9. Repeat the same steps at the other end using the same T568B order (this makes a straight-through cable).
10. Plug both ends into the cable tester. All 8 lights should glow one by one in order from 1 to 8.

Two common mistakes to avoid

- Wrong colour order: Mixing up the colours, or using T568A on one end and T568B on the other by mistake, gives a wrong cable (crossover) or no connection at all. Always check the order before crimping.
- Wires not fully inserted, or too much untwisting: If the wires do not reach the end of the connector, the pins do not touch the copper and the connection fails. Untwisting the pairs more than about 1.3 cm also increases crosstalk and reduces speed.

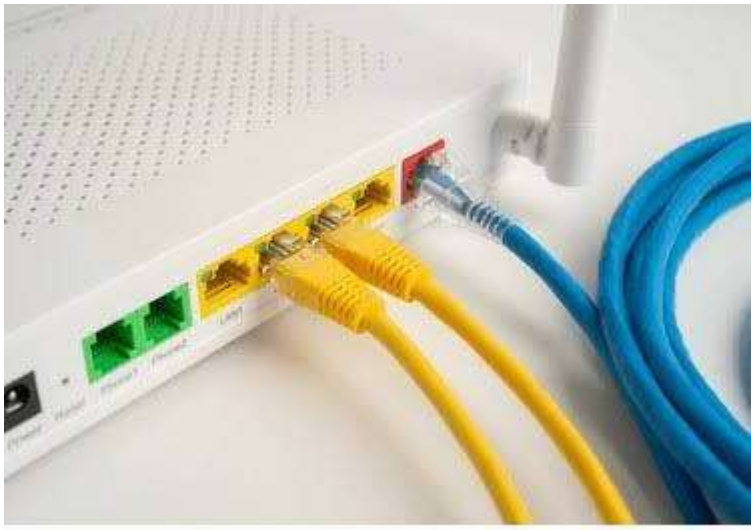
Q4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Connection	Cable	Connector	Reason
Consoles / gaming PCs / smart screens to the switch	Cat6 (or Cat6a) UTP patch cable	RJ-45	Gives stable 1 Gbps with low delay and less interference, and the cable is cheap enough to use for many devices.
Switch to the main router (and between switches)	Cat6a, or fiber optic if the distance is long	RJ-45 (Cat6a), or LC (fiber)	Carries the combined traffic of all devices without becoming slow, and fiber also avoids interference from many electrical devices in the hall.
Router to the internet (ISP)	Fiber optic	LC / SC (supplied by ISP)	Gives the high internet speed needed for live IPL streaming and online games.
Screens to the consoles / streaming box (display)	HDMI cable	HDMI	Used for the video and sound to the screen; this is separate from the network cable.

Reasons for the choice

- Cat6 with RJ-45 is reliable, fast and cost-effective for a large number of devices, and every device gets a wired connection, which is more stable than WiFi when many people are present.
- A gigabit managed switch (for example 24 ports) connects all devices in a star layout, so one faulty cable affects only one device.
- Fiber is used for long or high-capacity links because it carries more data over longer distances and is not affected by electrical noise.
- Good practice: keep each copper cable under 100 m, use ready-made tested cables, label every cable, and keep network cables away from power cables to avoid interference.

Q5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects. Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Type: A Cat6 UTP patch cable (straight-through, T568B) with an RJ-45 connector at each end.

Where it is used: Behind the WiFi router, plugged into one of the LAN ports.

What it connects: The WiFi router's LAN port to a laptop/desktop PC (or to a switch). A separate cable from the ISP connects to the router's WAN/internet port.

Description: The cable has a clear plastic RJ-45 plug with 8 gold-coloured pins and a small latch clip that locks it into the port. It carries data between the router and the computer at up to 1 Gbps, and it is a wired link that is more stable than WiFi.